Supplementary Information

Climate Controls on Runoff Across Hydroclimatic Regimes Using ERA5-Land and Machine Learning Analysis

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S1 Seasonal trend structure

Sen's slope magnitudes for all four variables at annual and seasonal resolution are shown in Fig. S1. The heatmap complements Table 2 of the main text by exposing the seasonal breakdown of the warming signal and the seasonal structure of Italian evapotranspiration increase. Italy's *ET* increase is concentrated in spring (MAM, +2.0 mm yr^{-1}) and summer (JJA, +1.4 mm yr^{-1}); Bangladesh's precipitation decline is distributed across all seasons but is strongest in winter (DJF) and autumn (SON). Saudi warming is largest in autumn and winter, a pattern consistent with enhanced night-time temperature increases in arid regions.

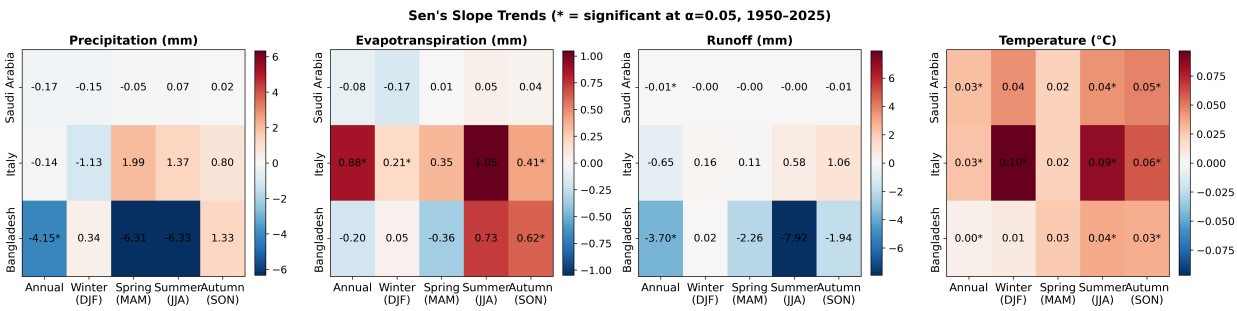

Figure S1: Mann–Kendall/Sen's slope trend magnitudes for precipitation, evapotranspiration, runoff, and temperature, by country and season (1950–2025). Asterisks denote statistical significance at $\alpha = 0.05$.

S2 Full annual SHAP feature distributions

The bar chart in Fig. 5 of the main text summarises SHAP contributions by their mean absolute value, but hides the direction and spread of individual contributions. Figure S2 shows the full SHAP distribution for every feature using beeswarm plots. The direction encoding confirms that high precipitation values consistently drive positive SHAP contributions (red points on the positive axis), while high soil temperature values contribute negatively to predicted runoff in Italy (red points on the negative axis), consistent with evapotranspiration suppression under warm conditions.

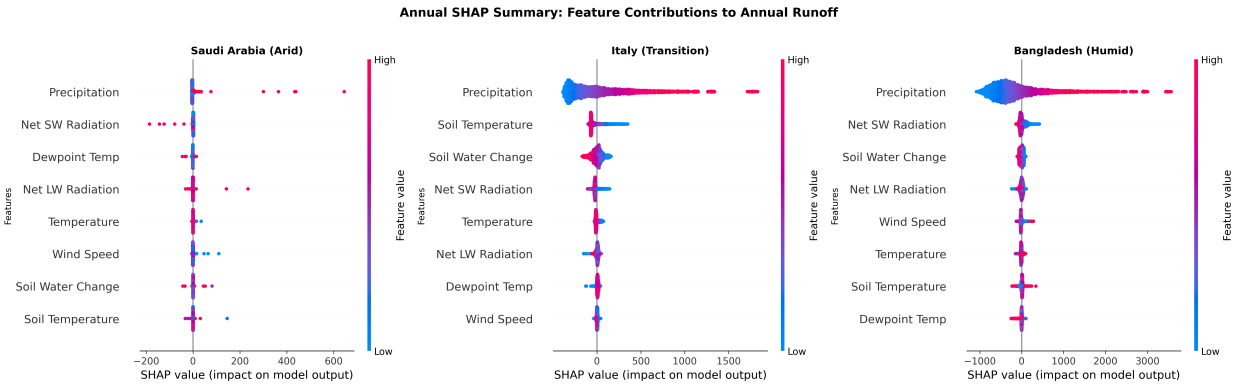

Figure S2: Annual SHAP beeswarm plots. Each point is one pixel-year from the test set. Horizontal position: SHAP contribution to predicted runoff; colour: raw feature value (red high, blue low).

S3 Temporal evolution of SHAP attributions

Figure S3 shows how the mean annual SHAP contribution of each feature has evolved between 1950 and 2025. Year-to-year variability is dominated by precipitation, particularly in Bangladesh, where the 1970s to 1990s show repeated high-SHAP excursions coinciding with strong monsoon years. In Italy, a mild declining trend in the precipitation SHAP contribution during the post-2000 period is consistent with the parallel increase in evapotranspiration documented in the main text (Section 4.2). Saudi Arabia shows no coherent temporal structure in any feature beyond the first decade, consistent with the low overall predictability of the arid-zone model.

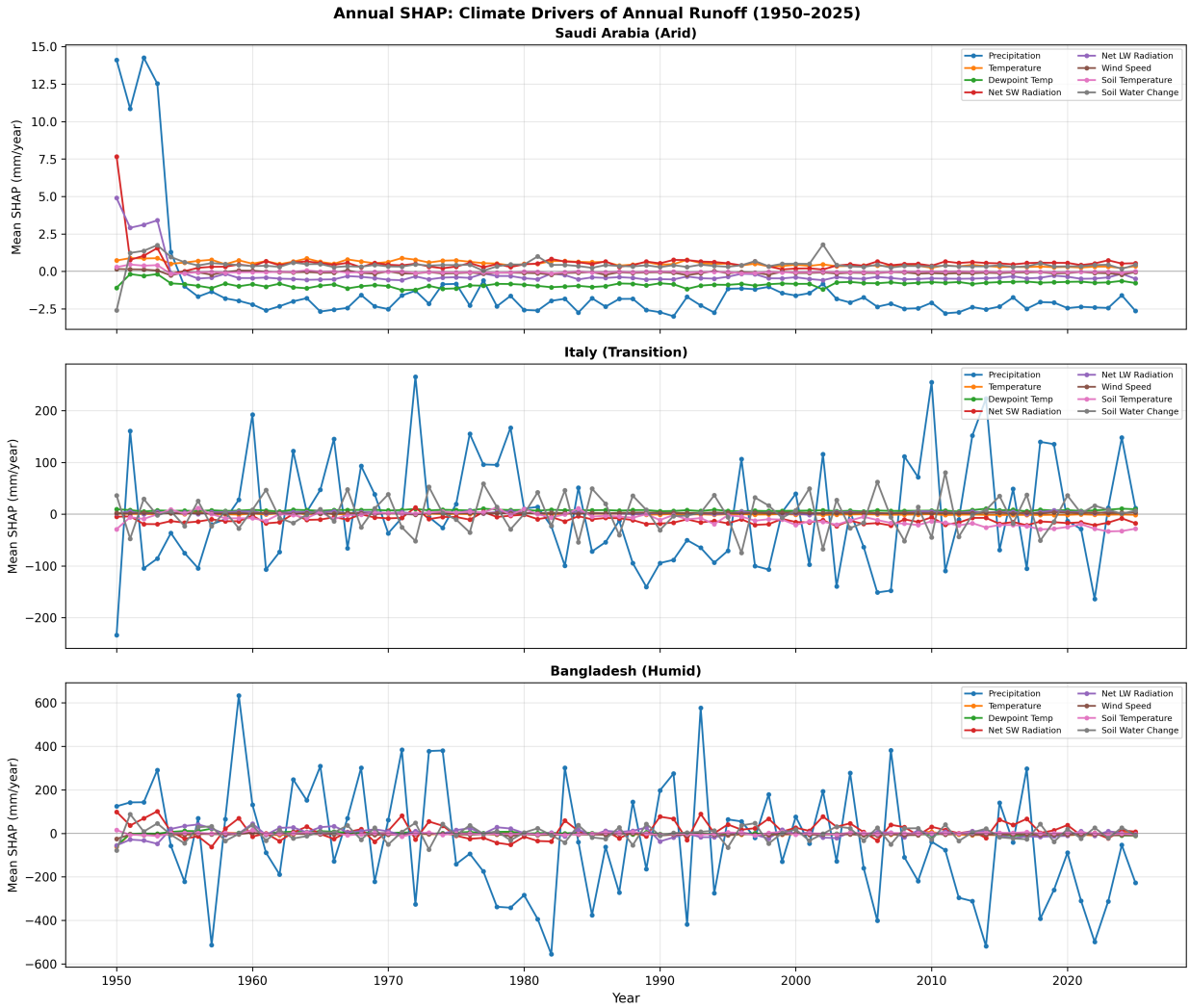

Figure S3: Annual mean SHAP contribution per feature, 1950–2025, for each country. One marker per calendar year, no smoothing applied.

S4 Monthly-scale models: justifying the annual focus

In addition to the annual model used in the main text, we also trained parallel monthly XGBoost models on pixel-month records. Their performance and SHAP rankings motivate why we chose the annual model for physical interpretation in the main text.

The monthly model performs markedly worse than the annual model in the hyper-arid regime ($R^2 = 0.36$ vs. 0.59 for Saudi Arabia). It also exhibits a seasonality-driven attribution anomaly: in Italy the monthly model ranks soil temperature as the top feature (mean |SHAP| $\sim$56 mm month^{-1}), well above precipitation ($\sim$16 mm month^{-1}). We interpret monthly soil temperature as a seasonal proxy — low winter values coincide with high runoff from snowmelt and low evapotranspiration, while high summer values coincide with suppressed runoff — and the model exploits this seasonal covariance as a predictive shortcut. Aggregation to annual resolution removes this within-year covariance and places precipitation unambiguously first in all three countries. Monthly SHAP results are therefore informative about the model's internal structure but should not be used as a basis for physical attribution, which is why the main text reports only annual results.

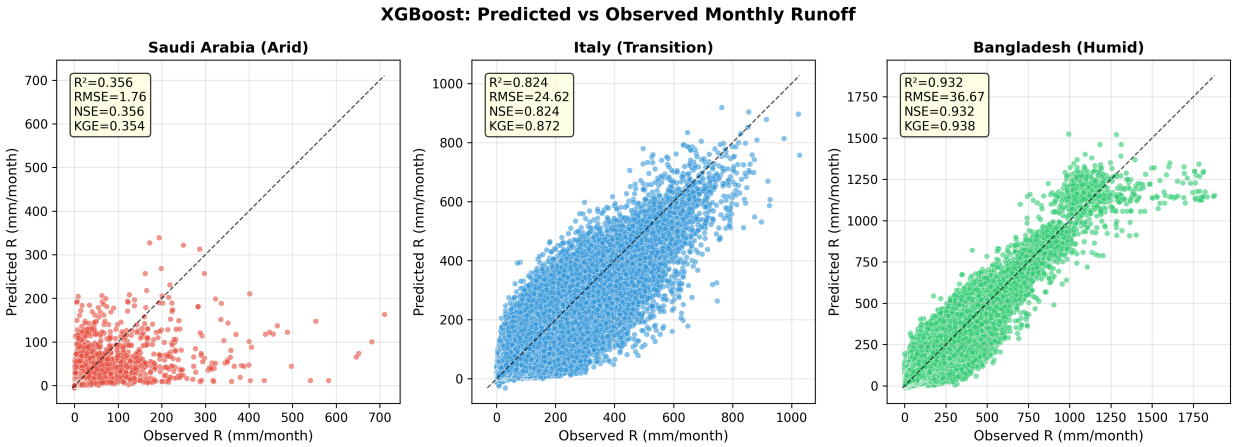

Figure S4: Monthly XGBoost predictions versus ERA5-Land observed runoff on the held-out test period (2005–2025). Compare with Fig. 4 of the main text (annual model).

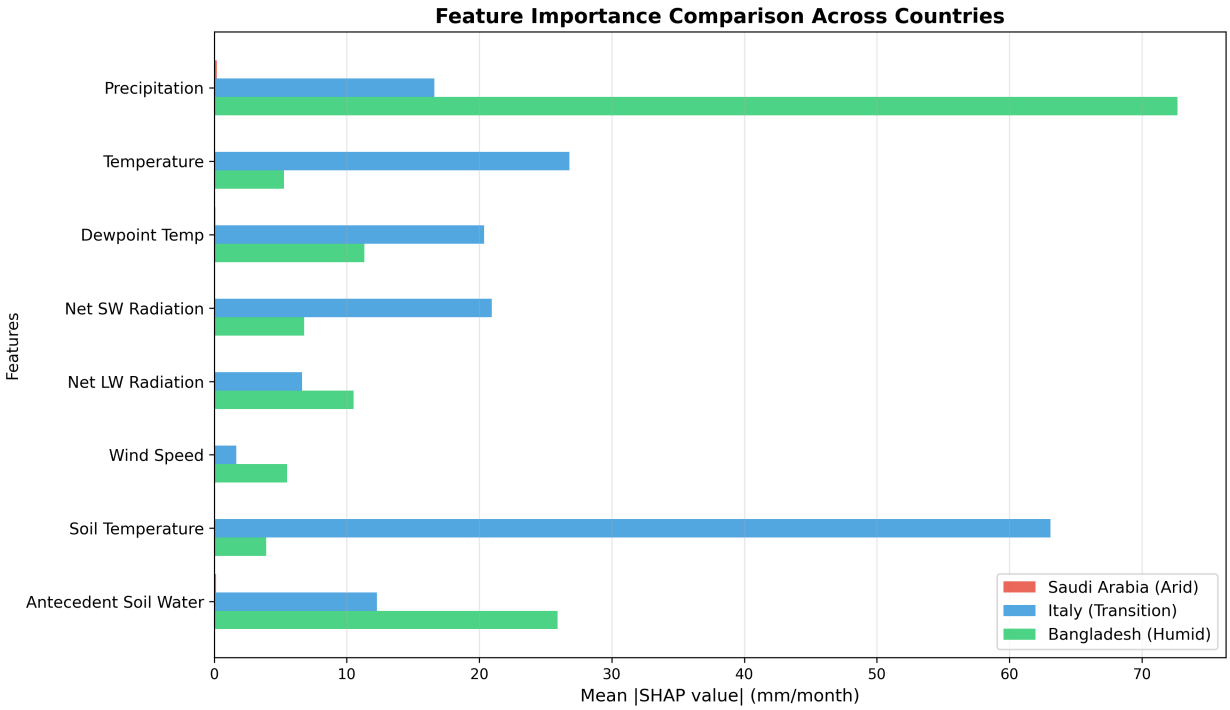

Figure S5: Monthly-model mean absolute SHAP values. Note that soil temperature exceeds precipitation in Italy, an artefact of strong seasonal covariance at the monthly scale.